

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Assignment

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Q5

Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q1

Banking Transaction System – Exception
Handling10
marks

Develop a Java program for an ATM withdrawal system.
The program accepts the account balance and
withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario	Expected Result	Purpose
10000 3000	Normal case	
10000 12000	Boundary / exception case	
10000 -500	Invalid input	
abc 3000	Invalid input	

Your Code

```
1 import java.util.*;
2
3 public class Main{
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6
7         try{
8             double balance = sc.nextDouble()
9             double withDrawalAmount = sc.next
10
11             if(withDrawalAmount <= 0){
12                 System.out.println("Invalid
13                 return;
14             }
15             if(withDrawalAmount > balance){
16                 throw new InsufficientBalanc
17             }
18
19             double remainingBalance = balanc
20             System.out.println("Transaction
21
22         }catch(InputMismatchException e){
23             System.out.println("Invalid input
24         }catch(InsufficientBalanceException
25             System.out.println(e.getMessage(
26         }finally{
27             System.out.println("Transaction
28             sc.close();
29         }
30     }
31 }
32
33 class InsufficientBalanceException extends E
34     public InsufficientBalanceException(Stri
35         super(msg);
36     }
37 }
```



Input

10000

3000



Output

```
Transaction completed. Remaining Balance: 7000  
Transaction finished
```

✓ Submitted

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